SHIFENG(SHI-FENG) SUN

Associate Professor
School of Computer Science
Shanghai Jiao Tong University, China

EDUCATION

Shanghai Jiao Tong University *2011/09 - 2016/12*
Phd in Computer Science & Engineering
Supervisor: Prof. Dawu Gu

The University of Melbourne *2015/04 - 2016/07*
Phd in Computer Science & Engineering (Joint Phd Student)
Supervisor: Prof. Udaya Parampalli

RESEARCH INTERESTS

Applied Cryptography, Encrypted Database, and Privacy-Enhancing Technologies

WORK EXPERIENCE

Assoc. Prof. at Shanghai Jiao Tong University, China *2021/09 - present*

Lecturer at Monash University, Australia *2020/04 - 2021/09*

Research Fellow at Monash University, Australia *2017/08 - 2020/03*

Postdoc Fellow at Hong Kong Polytechnic University, China *2017/05 - 2017/08*

ACADEMIC SERVICES

Publication Chair: ProvSec 2025, ASIACCS 2023, ACISP 2020, ISPEC 2019

Program Committee: ACM CCS 2025, ASIACRYPT 2023, ICICS 2023, NSS 2023, ProvSec 2023, ICISC 2019-2021, ACISP 2020

External Reviewers: CRYPTO 2018-2019, EUROFRYPT 2019, ASIACRYPT 2019-2022, TCC 2019, PKC 2019, IEEE TDSC, IEEE TIFS, IEEE TSC, etc.

PUBLICATIONS

Shifeng has published over 80 quality papers in cryptography and cybersecurity, including the publications in top-tier venues like **ACM CCS**, **NDSS**, **USENIX Security**, **EUROCRYPT**, **ASIACRYPT**, **ESORICS**, **PKC**, **IEEE TDSC**, **IEEE TKDE**, and **IEEE TSC**. The selected publications are as follows:

50. Xinzhou Wang, **Shi-Feng Sun**, Rupeng Yang, Junqing Gong, Dawu Gu, Yuan Luo. Threshold Homomorphic Secret Sharing: Definitions and Constructions. To appear at ASIACRYPT 2025.

49. Guowei Ling, Peng Tang, **Shi-Feng Sun**, Shouling Ji, Wedong Qiu. Ultra-Fast Private Set Intersection from Efficient Oblivious Key-Value Stores. IEEE Transactions on Dependable and Secure Computing, 2025.

48. Lei Zhang, Jianfeng Wang, Jun Wu, Yunling Wang, **Shi-Feng Sun**. Violin: Powerful Volumetric Injection Attack Against Searchable Encryption With Optimal Injection Size. IEEE Transactions on Dependable and Secure Computing, 2025.

47. Cong Zuo, Shangqi Lai, **Shi-Feng Sun**, Xingliang Yuan, Joseph K. Liu, Jun Shao, Huaxiong Wang, Liehuang Zhu, Shujie Cui. Searchable Encryption for Conjunctive Queries with Extended Forward and Backward Privacy. Proc. Priv. Enhancing Technol. 2025(1): 440-455 (2025).

46. Qiaoer Xu, Jianfeng Wang, **Shi-Feng Sun**, Zhipeng Liu, Xiaofeng Chen. Practical Equi-Join Over Encrypted Database With Reduced Leakage. IEEE Trans. Knowl. Data Eng. 37(5): 2846-2860.

45. Chunyang Lv, Jianfeng Wang, **Shi-Feng Sun**, Yunling Wang, Saiyu Qi, Xiaofeng Chen. Towards Practical Multi-Client Order-Revealing Encryption: Improvement and Application. IEEE Trans. Dependable Secur. Comput. 21(3): 1111-1126 (2024).

44. Saiyu Qi, Wei Wei, Jianfeng Wang, **Shi-Feng Sun**, Leszek Rutkowski, Tingwen Huang, Janusz Kacprzyk, Yong Qi. Secure Data Deduplication With Dynamic Access Control for Mobile Cloud Storage. IEEE Trans. Mob. Comput. 23(4): 2566-2582 (2024).

43. Gongyu Shi, Geng Wang, **Shi-Feng Sun***, Dawu Gu. Efficient cryptanalysis of an encrypted database supporting data interoperability. VLDB J. 33(5): 1357-1375 (2024).

42. Zhilong Luo, **Shi-Feng Sun***, Zhedong Wang, Dawu Gu. Non-interactive Publicly Verifiable Searchable Encryption with Forward and Backward Privacy. ACISP 2024: 281-302.

41. Chenke Wang, Zhonghui Ge, Yu Long, Xian Xu, Shi-Feng Sun, Dawu Gu. BlindShuffler: Universal and Trustless Mixing for Confidential Transactions. AsiaCCS 2024.

40. Yunling Wang, **Shi-Feng Sun***, Jianfeng Wang, Xiaofeng Chen, Joseph K. Liu, Dawu Gu. Practical Non-interactive Encrypted Conjunctive Search with Leakage Suppression. ACM CCS 2024: 4658-4672.

39. Yuncong Zhang, **Shi-Feng Sun***, Dawu Gu. Efficient KZG-Based Univariate Sum-Check and Lookup Argument. Public Key Cryptography 2024: 400-425.

38. Boshi Yuan, Shixuan Yang, Yongxiang Zhang, Ning Ding, Dawu Gu, **Shi-Feng Sun***. MD-ML: Super Fast Privacy-Preserving Machine Learning for Malicious Security with a Dishonest Majority. USENIX Security Symposium 2024.

37. **Shi-Feng Sun***, Ron Steinfeld, Amin Sakzad. Incremental symmetric puncturable encryption with support for unbounded number of punctures. Des. Codes Cryptogr. 91(4): 1401-1426.

36. Yanli Ren, Zhuhuan Song, **Shi-Feng Sun***, Joseph K. Liu, Guorui Feng. Outsourcing LDA-Based Face Recognition to an Untrusted Cloud. IEEE Trans. Dependable Secur. Comput. 20(3): 2058-2070 (2023).

35. Mingyun Bian, Joseph K. Liu, **Shi-Feng Sun**, Xinpeng Zhang, Yanli Ren. Verifiable Privacy-Enhanced Rotation Invariant LBP Feature Extraction in Fog Computing. IEEE Trans. Ind. Informatics 19(12): 11518-11530 (2023).

34. Yuncong Zhang, **Shi-Feng Sun***, Ren Zhang, Dawu Gu. Polynomial IOPs for Memory Consistency Checks in Zero-Knowledge Virtual Machines. ASIACRYPT 2023: 111-141.

33. Geng Wang, **Shi-Feng Sun**, Zhedong Wang, Dawu Gu. Functional Encryption Against Probabilistic Queries: Definition, Construction and Applications. Public Key Cryptography 2023: 429-458.

32. Viet Vo, Xingliang Yuan, **Shi-Feng Sun**, Joseph K. Liu, Surya Nepal, Cong Wang. ShieldDB: An Encrypted Document Database With Padding Countermeasures. IEEE Trans. Knowl. Data Eng. 35(4): 4236-4252 (2023)

31. Yanxue Jia, **Shi-Feng Sun***, Hong-Sheng Zhou, Jiajun Du, Dawu Gu. Shuffle-based Private Set Union: Faster and More Secure. USENIX Security 2022: 2947-2964.

30. Yanxue Jia, **Shi-Feng Sun**[*], Hong-Sheng Zhou, Dawu Gu. A Universally Composable Non-interactive Aggregate Cash System. ASIACRYPT 2022: 745-773.

29. Jianfeng Wang, **Shi-Feng Sun**[†], Tianci Li, Saiyu Qi, Xiaofeng Chen. Practical Volume-Hiding Encrypted Multi-Maps with Optimal Overhead and Beyond. ACM CCS 2022: 2825-2839.

28. Yuncong Zhang, Alan Szepeniec, Ren Zhang, **Shi-Feng Sun**[*], Geng Wang, Dawu Gu. VOProof: Efficient zkSNARKs from Vector Oracle Compilers. ACM CCS 2022: 3195-3208.

27. **Shi-Feng Sun**, Ron Steinfeld, Shangqi Lai, Xingliang Yuan, Amin Sakzad, Joseph K. Liu, Surya Nepal, Dawu Gu. Practical Non-Interactive Searchable Encryption with Forward and Backward Privacy. The Network and Distributed System Security Symposium (NDSS) 2021.

26. **Shi-Feng Sun**, Amin Sakzad, Ron Steinfeld, Joseph K. Liu, Dawu Gu. Public-Key Puncturable Encryption: Modular and Compact Constructions. Public Key Cryptography (PKC) 2020: 309-338.

25. Veronika Kuchta, Amin Sakzad, Damien Stehlé, Ron Steinfeld, **Shi-Feng Sun**. Measure-Rewind-Measure: Tighter Quantum Random Oracle Model Proofs for One-Way to Hiding and CCA Security. EUROCRYPT (3) 2020: 703-728.

24. Cong Zuo, **Shi-Feng Sun**[*], Joseph K. Liu, Jun Shao, Josef Pieprzyk, Lei Xu. Forward and Backward Private DSSE for Range Queries. IEEE Transactions on Dependable and Secure Computing, doi: 10.1109/TDSC.2020.2994377.

23. Viet Vo, Shangqi Lai, Xingliang Yuan, **Shi-Feng Sun**, Surya Nepal, Joseph K. Liu: Accelerating Forward and Backward Private Searchable Encryption Using Trusted Execution. ACNS 2020

22. Yanxue Jia, **Shi-Feng Sun**[*], Yuncong Zhang, Qingzhao Zhang, Ning Ding, et al. PBT: A New Privacy-Preserving Payment Protocol for Blockchain Transactions. IEEE Transactions on Dependable and Secure Computing, doi: 10.1109/TDSC.2020.2998682.

21. Shabnam Kasra Kermanshahi, **Shi-Feng Sun**, Joseph K. Liu, Ron Steinfeld, et al. Geometric range search on encrypted data with Forward/Backward security. IEEE Transactions on Dependable and Secure Computing, doi: 10.1109/TDSC.2020.2982389.

20. Tsz Hon Yuen, **Shi-Feng Sun**, Joseph K. Liu, Man Ho Au, Muhammed F. Esgin, Qingzhao Zhang, Dawu Gu: RingCT 3.0 for Blockchain Confidential Transaction: Shorter Size and Stronger Security. Financial Cryptography 2020: 464-483.

19. **Shi-Feng Sun**, Cong Zuo, Joseph K. Liu, et al. Non-Interactive Multi-Client Searchable Encryption: Realization and Implementation. IEEE Transactions on Dependable and Secure Computing, doi: 10.1109/TDSC.2020.2973633.

18. Yunling Wang, **Shi-Feng Sun**, Jianfeng Wang, Joseph K. Liu, Xiaofeng Chen. Achieving Searchable Encryption Scheme with Search Pattern Hidden. IEEE Transactions on Services Computing, doi: 10.1109/TSC.2020.2973139.

17. Cong Zuo, **Shi-Feng Sun**[*], Joseph K. Liu, Jun Shao and Josef Pieprzyk. Dynamic Searchable Symmetric Encryption with Forward and Stronger Backward Privacy. ESORICS (2) 2019: 283-303.

16. Maxime Buser, Joseph K. Liu, Ron Steinfeld, Amin Sakzad and **Shi-Feng Sun**. A Dynamic & Revocable Group Merkle Signature. ESORICS (1) 2019: 194-214.

15. **Shi-Feng Sun**, Dawu Gu, Man Ho Au, Shuai Han, Yu Yu, Joseph K. Liu. Strong Leakage and Tamper-Resilient Public Key Encryption from Refined Hash Proof System. ACNS 2019: 486-506.

14. Shangqi Lai, Xingliang Yuan, **Shi-Feng Sun**, Joseph K. Liu, Yuhong Liu, Dongxi Liu. GraphSE^2: An Encrypted Graph Database for Privacy-Preserving Social Search. AsiaCCS 2019: 41-54.

13. Lei Xu, **Shi-Feng Sun**, Xingliang Yuan, Joseph K. Liu, Cong Zuo, Chungen Xu. Enabling Authorized Encrypted Search for Multi-Authority Medical Databases. IEEE Transactions on Emerging Topics in Computing. DOI: 10.1109/TETC.2019.2905572
12. **Shi-Feng Sun**, Xingliang Yuan, Joseph K. Liu, Ron Steinfeld, Amin Sakzad, Viet Vo, Surya Nepal. Practical Backward-Secure Searchable Encryption from Symmetric Puncturable Encryption. ACM CCS 2018: 763-780.
11. Shangqi Lai, Sikhar Patranabis, Amin Sakzad, Joseph K. Liu, Debdeep Mukhopadhyay, Ron Steinfeld, **Shi-Feng Sun***, Dongxi Liu, Cong Zuo. Result Pattern Hiding Searchable Encryption for Conjunctive Queries. ACM CCS 2018: 745-762.
10. Cong Zuo, **Shi-Feng Sun***, Joseph K. Liu, Jun Shao, Josef Pieprzyk. Dynamic Searchable Symmetric Encryption Schemes Supporting Range Queries with Forward (and Backward) Security. ESORICS (2) 2018: 228-246.
9. Jianfeng Wang, Xiaofeng Chen, **Shi-Feng Sun**, Joseph K. Liu, Man Ho Au and Zhi-Hui Zhan. Towards Efficient Verifiable Conjunctive Keyword Search for Large Encrypted Database. ESORICS (2) 2018: 83-100.
8. Yunling Wang, Jianfeng Wang, **Shi-Feng Sun**, Joseph K. Liu, Willy Susilo and Xiaofeng Chen. Towards Multi-user Searchable Encryption Supporting Boolean Query and Fast Decryption. ProvSec 2017: 24-38. (EI)
7. **Shi-Feng Sun**, Man Ho Au, Joseph K. Liu, Tsz Hon Yuen. RingCT 2.0: A Compact Accumulator-Based (Linkable Ring Signature) Protocol for Blockchain Cryptocurrency Monero. ESORICS (2) 2017: 456-474.
6. **Shi-Feng Sun**, Dawu Gu, Udaya Parampalli, et al. Public Key Encryption Resilient to Leakage and Tampering Attacks. J. Comput. Syst. Sci. 89: 142-156 (2017).
5. Baodong Qin, Shengli Liu, **Shi-Feng Sun**, Robert H. Deng, Dawu Gu. Related-key Secure Key Encapsulation from Extended Computational Bilinear Diffie-Hellman. Inf. Sci. 406: 1-11 (2017).
4. **Shi-Feng Sun**, Joseph K. Liu, Amin Sakzad, et al. An Efficient Non-Interactive Multi-Client Searchable Encryption with Support for Boolean Queries. In proceedings of 21st European Symposium on Research in Computer Security, ESORICS '16, Heraklion, Greece, September 26-30, 2016, pp. 154–172.
3. **Shi-Feng Sun**, Udaya Parampalli, Tsz Hon Yuen, Yu Yu, and Dawu Gu. Efficient Completely Non-Malleable and RKA Secure Public Key Encryptions. In Proceedings of Information Security and Privacy - 21st Australasian Conference, ACISP'16, Melbourne, VIC, Australia, July 4-6, 2016, pp. 134-150.
2. **Shi-Feng Sun**, Joseph K. Liu, Yu Yu, et al. RKA-Secure Public Key Encryptions against Efficiently Invertible Functions. The Computer Journal (2016) doi: 10.1093/comjnl/bxw025.
1. **Shi-Feng Sun**, Dawu Gu, Joseph K. Liu, et al. Efficient Construction of Completely Non-Malleable Public Key Encryption. In Proceedings of the 11th ACM on Asia Conference on Computer and Communications Security (ASIA CCS '16), pp. 901-906.

AWARDS & HONORS

IEEE TEMS TC Mid-Career Award, 2024

ACM China Council (Shanghai Chapter) Rising Star, 2023

Excellent Doctoral Dissertation Award, 2019